

Equation Match Sprint

Unit 7: Equation Detective Agency · Lesson 7-3 · EduWonderLab

UNIT 7 · LESSON 7-3 · TEACHER NOTES

Standard 6.EE.7

FORMAT Speed-match game

TIME 25–30 min

GROUPING Pairs

TOPIC Solve Multiplication and Division Equations

STANDARD 6.EE.7

PREP LEVEL Light — print equation cards and solution cards

Learning goal *I can solve multiplication and division equations using inverse operations.*

Teacher setup

- Print one set per pair. Students solve each equation and match it to a solution card.
- Partners agree on the inverse operation before matching.
- Use the integer set first, then the challenge set.

Materials

Equation cards, Solution cards, Inverse operation organizer, Pencil.

Differentiation & language support

- Organizer: $\times$ undoes $\div$, $\div$ undoes $\times$ (SPED).
- Sentence stem: “I divided both sides by ___ to undo the ___.”
- ESOL: show the operation that is happening to the variable.

Key vocabulary (English / Español):

Term	Español	What it means
coefficient	coeficiente	the number multiplied by a variable
inverse operation	operación inversa	the operation that undoes another

Quick teacher check: *Ask a pair to name the inverse operation used on Card 5.*

Equation Match Sprint

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Name: _____

Date: _____

Class: _____

Your goal: *I can solve multiplication and division equations using inverse operations.*

What you need

Equation cards, Solution cards.

How to play

1. Pick an equation card.
2. Decide the inverse operation with your partner.
3. Solve and write the solution.
4. Match it to the correct solution card.

Equation Cards

Undo multiplication with division, and division with multiplication. Do it to both sides.

1. $3x = 12$	2. $5x = 35$
3. $x \div 4 = 6$	4. $2x = 18$
5. $x \div 3 = 5$	6. $6x = 42$
7. $x \div 2 = 9$	8. $8x = 56$
9. $4x = 10$	10. $x \div 5 = 3$

Record your work (inverse operation + solution):

Explain your thinking

Explain why you divide both sides by the coefficient to solve $3x = 12$.

Sentence frame: *I ___ both sides by ___ to undo the ___.*

★ **Challenge:** Cards 2, 6, and 8 all give $x = 7$. Write a new equation with the solution $x = 7$.

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. $3x = 12$ Answer: $x = 4$	2. $5x = 35$ Answer: $x = 7$
3. $x \div 4 = 6$ Answer: $x = 24$	4. $2x = 18$ Answer: $x = 9$
5. $x \div 3 = 5$ Answer: $x = 15$	6. $6x = 42$ Answer: $x = 7$
7. $x \div 2 = 9$ Answer: $x = 18$	8. $8x = 56$ Answer: $x = 7$
9. $4x = 10$ Answer: $x = 2.5$	10. $x \div 5 = 3$ Answer: $x = 15$

Teaching notes & look-fors

- Multiplication equation → divide; division equation → multiply.
- Cards 2, 6, 8 all solve to $x = 7$.